

Lab 2 Report

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Code

Code for AIMS

```
Lab02 > Code > src > default_package > J Aims.java > Language Support for Java(TM) by Red Hat > Aims > main(String[])
You, 4 days ago | 1 author (You)
1 package default_package;
2
You, 4 days ago | 1 author (You)
3 public class Aims {
4
5     Run main | Debug main | Run | Debug
    public static void main(String[] args) {
6         Implement with Codex
7         // TODO Auto-generated method stub
8         //Create a new cart
9         Cart anOrder = new Cart();
10
11         You, 4 days ago * "Update Lab02"
12         //Create new dvd objects and add them to the cart
13         DigitalVideoDisc dvd1 = new DigitalVideoDisc("The Lion King",
14             "Animation", "Roger Allers", 87, 19.95f);
15         anOrder.addDigitalVideoDisc(dvd1);
16         DigitalVideoDisc dvd2 = new DigitalVideoDisc("Star Wars",
17             "Science Fiction", "George Lucas", 87, 24.95f);
18         anOrder.addDigitalVideoDisc(dvd2);
19         DigitalVideoDisc dvd3 = new DigitalVideoDisc("Aladin",
20             "Animation", 18.99f);
21         anOrder.addDigitalVideoDisc(dvd3);
22
23         //print out total cost of the items in the cart
24         System.out.print("Total Cost is: ");
25         System.out.println(anOrder.totalCost());
26
27         //test out remove method
28         anOrder.removeDigitalVideoDisc(dvd3);
29         System.out.print("Total Cost is: ");
30         System.out.println(anOrder.totalCost());
31
32     }
33 }
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS

(base) Hieu-MacBook-Pro:00P_Lab hieunguyen\$ java /Users/hieunguyen/00P_Lab/Lab02/Code/src/default_package/Aims.java

The disc has been added to the cart.
The disc has been added to the cart.
Total Cost is: 63.89
The disc has been removed from the cart.
Total Cost is: 44.9

(base) Hieu-MacBook-Pro:00P_Lab hieunguyen\$

Indexing completed. Java: Warning DVC (Auto) You, 4 days ago Hieu (4 days ago) Ln 9, Col 17 Tab Size: 4 UTF-8 LF Java Signed out Miku

Code for Cart

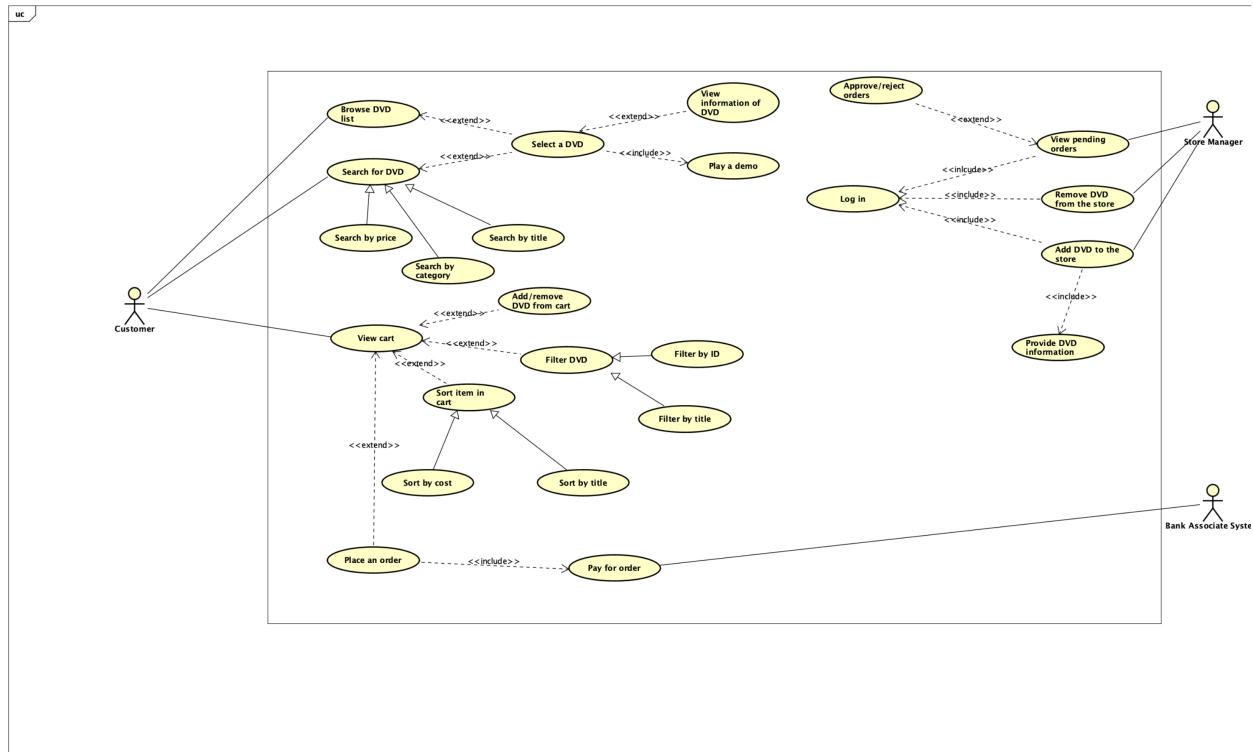
```
Lab02 > Code > src > default_package > J Cart.java > Language Support for Java(TM) by Red Hat > Cart > addDigitalVideoDisc(DigitalVideoDisc)

You, 4 days ago | 1 author (You)
1 package default_package;
2
3 You, 4 days ago | 1 author (You)
4 public class Cart {
5     public static final int MAX_NUMBERS_ORDERED=20;
6     private DigitalVideoDisc itemsOrdered[]=
7         new DigitalVideoDisc[MAX_NUMBERS_ORDERED];
8     private int qtyOrdered = 0;
9
10    public void addDigitalVideoDisc(DigitalVideoDisc disc) {
11        if (qtyOrdered >= MAX_NUMBERS_ORDERED) {
12            System.out.println("The cart is full! Cannot add more discs.");
13        } else {
14            itemsOrdered[qtyOrdered] = disc;
15            qtyOrdered++;
16            System.out.println("The disc has been added to the cart.");
17            if (qtyOrdered == MAX_NUMBERS_ORDERED - 1) {
18                System.out.println("The cart is almost full.");
19            }
20        }
21
22        You, 4 days ago • "Update Lab02"
23
24    public void removeDigitalVideoDisc(DigitalVideoDisc disc) {
25        boolean found = false;
26        for (int i = 0; i < qtyOrdered; i++) {
27            if (itemsOrdered[i].equals(disc)) {
28                found = true;
29                for (int j = i; j < qtyOrdered - 1; j++) {
30                    itemsOrdered[j] = itemsOrdered[j + 1];
31                }
32                itemsOrdered[qtyOrdered - 1] = null;
33                qtyOrdered--;
34                System.out.println("The disc has been removed from the cart.");
35                break;
36            }
37        }
38        if (!found) {
39            System.out.println("The disc is not found in the cart.");
40        }
41    }
42
43    public float totalCost() {
44        float total = 0;
45        for (int i = 0; i < qtyOrdered; i++) {
46            total += itemsOrdered[i].getCost();
47        }
48        return total;
49    }
50
51 PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GIT LENS
52 (base) Hieus-MacBook-Pro:00P_Lab hieunguyen$ java /Users/hieunguyen/00P_Lab/Lab02/Code/src/default_package/Aims.java
53 The disc has been added to the cart.
54 The disc has been added to the cart.
55 Total Cost is: 63.89
56 The disc has been removed from the cart.
57 Total Cost is: 44.9
58 (base) Hieus-MacBook-Pro:00P_Lab hieunguyen$
```

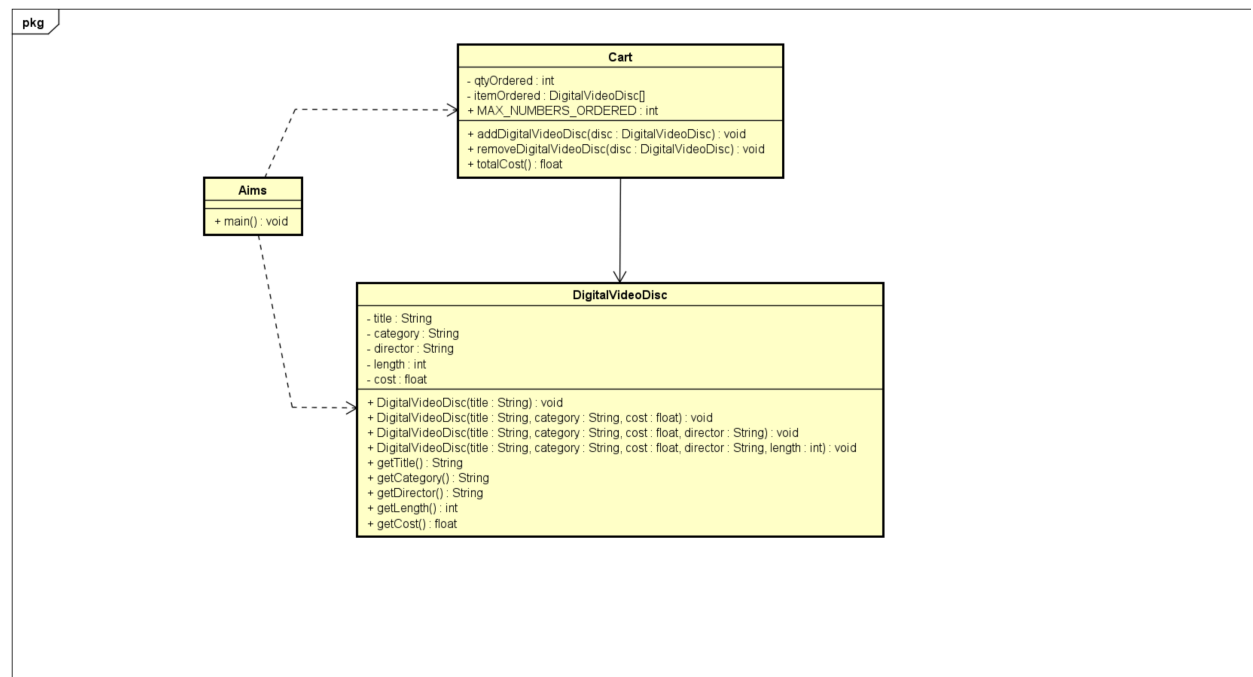
Code for DigitalVideoDisc

```
Lab02 > Code > src > default_package > J DigitalVideoDisc.java > Language Support for Java(TM) by Red Hat > {} default_package
You, 4 days ago | 1 author (You)
1 package default_package; You, 4 days ago * "Update Lab02" ...
2
You, 4 days ago | 1 author (You)
3 public class DigitalVideoDisc {
4     public String getTitle() {
5         return title;
6     }
7     public String getCategory() {
8         return category;
9     }
10    public String getDirector() {
11        return director;
12    }
13    public int getLength() {
14        return length;
15    }
16    public float getCost() {
17        return cost;
18    }
19    public DigitalVideoDisc(String title) {
20        super();
21        this.title = title;
22    }
23    public DigitalVideoDisc(String title, String category, float cost) {
24        super();
25        this.title = title;
26        this.category = category;
27        this.cost = cost;
28    }
29    public DigitalVideoDisc(String title, String category, String director, float cost) {
30        super();
31        this.title = title;
32        this.category = category;
33        this.director = director;
34        this.cost = cost;
35    }
36    public DigitalVideoDisc(String title, String category, String director, int length, float cost) {
37        super();
38        this.title = title;
39        this.category = category;
40        this.director = director;
41        this.length = length;
42        this.cost = cost;
43    }
44    private String title;
45    private String category;
46    private String director;
47    private int length;
48    private float cost;
49 }
50
51
```

Use Case Diagram



Class Diagram



Reading Assignment Mindmap

